



Complete Course Final Review

Certification Mock + Interview Simulation

Java 8+ Master Program



Session Structure

1. Full DSA recap
2. Core Java recap
3. Design patterns recap
4. Concurrency & JVM recap
5. System design mock
6. Certification test simulation
7. Final evaluation



Module 1: Core Java Review

Topics covered:

- OOPS & SOLID
- Collections Framework
- Streams & Functional Programming
- Concurrency
- JVM Internals
- Performance tuning



Quick Revision Questions

1. Difference between HashMap & ConcurrentHashMap?
2. How does GC work?
3. What is volatile?
4. Explain SOLID.
5. How Spring IoC works?



Module 2: Data Structures Recap

Covered:

- Arrays
- Strings
- Linked Lists
- Stack & Queue
- Hashing
- Trees & BST
- Heaps
- Graphs
- Segment Trees



Module 3: Algorithms Recap

Covered:

- Searching
- Sorting
- Recursion
- Backtracking
- Greedy
- Dynamic Programming
- Advanced graph algorithms



Coding Mock Round

Problem 1:

Longest Substring Without Repeating Characters.

Problem 2:

Implement Min Stack.

Problem 3:

Find Kth Largest using Heap.

Problem 4:

Detect Cycle in Directed Graph.

Each:

Explain time & space complexity.



System Design Mock

Design:

Online Food Delivery System.

Requirements:

- Order placement
- Restaurant search
- Real-time delivery tracking
- Payment integration
- High traffic handling

Expected Design Components

- API Gateway
- Microservices
- Database sharding
- Cache
- Load balancer
- Message queue
- Monitoring



Performance Review Questions

- How to reduce latency?
- How to scale horizontally?
- How to handle 10M users?
- How to prevent cascading failures?
- How to tune JVM?



Certification Mock Test (Part 1 — MCQs)

1. Time complexity of MergeSort?
2. What is Red-Black Tree?
3. Difference between Heap & Stack?
4. What is memoization?
5. What is eventual consistency?



Certification Mock Test (Part 2 — Short Answers)

1. Explain CAP theorem.
2. Describe difference between REST & SOAP.
3. How does thread pool improve performance?
4. What is optimistic locking?
5. Explain Builder pattern.



Certification Mock Test (Part 3 — Coding)

Implement:

1. LRU Cache
2. 0/1 Knapsack
3. Topological Sort
4. Binary Search (First Occurrence)



Evaluation Criteria

- Code clarity
- Correctness
- Complexity explanation
- Edge case handling
- Design reasoning
- Communication clarity



Real Interview Simulation

Flow:

15 min — DSA

30 min — Coding

30 min — System Design

20 min — Java Deep Dive

10 min — Behavioral

Common Final Mistakes

- Forgetting complexity explanation
- Ignoring edge cases
- Overcomplicating solution
- Poor communication
- Not thinking about scalability



Final Checklist

Before certification/interview:

- ✓ Understand all time complexities
- ✓ Practice binary search template
- ✓ Master DP state transitions
- ✓ Understand microservices architecture
- ✓ Know Spring internals
- ✓ Know concurrency concepts
- ✓ Practice explaining trade-offs



Real World Readiness

You can now:

- Solve advanced DSA problems
- Design scalable backend systems
- Write thread-safe Java code
- Tune JVM performance
- Architect microservices
- Handle senior-level interviews



Final Takeaway

This course built:

- Algorithmic thinking
- Architectural mindset
- Performance awareness
- Enterprise Java mastery
- Interview confidence

You are now equipped for:

Java Developer

Backend Engineer

System Design Interviews



Congratulations 🎓

You have completed:

Complete Java Master Program
From DSA to System Design

Keep practicing.

Keep building.

Keep optimizing.

Keep learning.

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